Quantum propagators and path integrals

Jianshu Cao

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I Quantum propagators

A Definition

$$\hat{G} = e^{-it\hat{H}} \qquad \begin{cases} i\partial_t \rho = \hat{H}\rho \\ \rho(t) = G\rho(0) \end{cases} \tag{1}$$

B Eigen-decomposition

eigen-energy ε_n

$$\hat{H}\phi_n = \lambda_n \phi_n \qquad \hat{H} = \sum_n \lambda_n \left|\phi_n\right\rangle\!\left\langle\phi_n\right| \tag{2}$$

$$\hat{G} = e^{-it\hat{H}} = e^{-it\lambda_n} \left|\phi_n \lambda\right\rangle\!\left\langle\phi_n\right| \tag{3}$$

C Split operator

$$\hat{H} = \hat{H}_0 + \Delta\hat{H} \qquad \hat{H}_0 \text{ is the reference} \tag{4}$$

$$t_f = N\delta t \qquad t_n = n\delta t \qquad n = \{0, N\} \tag{5}$$

$$\hat{G} = \left(e^{-i\hat{H}\delta t}\right)^N \tag{6}$$

Trotter expansion $e^{-i\hat{H}\delta t} \approx e^{-i\hat{H}_0\delta t} e^{-i\Delta\hat{H}\delta t}$

$$\hat{G} = \left(e^{-i\hat{H}_0\delta t} e^{-i\Delta\hat{H}\delta t}\right)^N + NO\left(\delta t^2\right) \tag{7}$$

D An example

$$\hat{H} = \underbrace{\frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2}_{H_0} + \underbrace{cx^4}_{\Delta H} \tag{8}$$

Let's adopt the harmonic basis

$$\hat{H}_0 = \hbar\omega\left(n + \frac{1}{2}\right)|n\rangle\langle n| \qquad n = \phi_n \tag{9}$$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}\left(a + a^\dagger\right) = \sqrt{\frac{\hbar}{2m\omega}}\sqrt{n}\left[|n\rangle\langle n-1| + |n-1\rangle\langle n|\right] \tag{10}$$

$$\Delta\hat{H} = c\hat{x}^4 \tag{11}$$

$$\hat{G}_{nm} = e^{-i\delta t\hbar\omega\left(n+\frac{1}{2}\right)}|n\rangle\langle n|e^{-i\delta t\Delta H}|m\rangle\langle m| \tag{12}$$

question: a better preference?

$$H = \underbrace{\frac{\hat{p}^2}{2m_1} + \frac{1}{2}m\bar{\omega}^2 x^2}_{H_{\rm ref}} + \underbrace{\frac{1}{2}m\left(\omega^2 - \bar{\omega}^2\right)x^2 + cx^4}_{\Delta H} \tag{13}$$

$$\langle \Delta H \rangle_{\rm ref} = 0 \tag{14}$$

$$\text{or} \begin{cases} \langle \Delta H' \rangle_{\rm ref} = 0 \\ \langle \Delta H'' \rangle_{\rm ref} = 0 \end{cases} \tag{15}$$

E Better Trotterization

Baker–Campbell–Hausdorff (BCH) Formula

$$e^B e^A = \exp\left(A + B + \frac{1}{2}[A,B] + \frac{1}{12}[A-B,[A,B]] + \cdots\right) \tag{16}$$

So,

$$\hat{G}(t) = \left(e^{-i\delta t H_0} e^{-i\delta \Delta H}\right)^N + NO\left(\delta t^2\right).$$

Alternatively,

$$\hat{G}(t) = \left[e^{-i\delta\Delta H/2} e^{-i\delta H_0} e^{-i\delta \Delta H/2}\right]^N + NO\left(\delta t^3\right) \tag{17}$$

Symmetric ordering

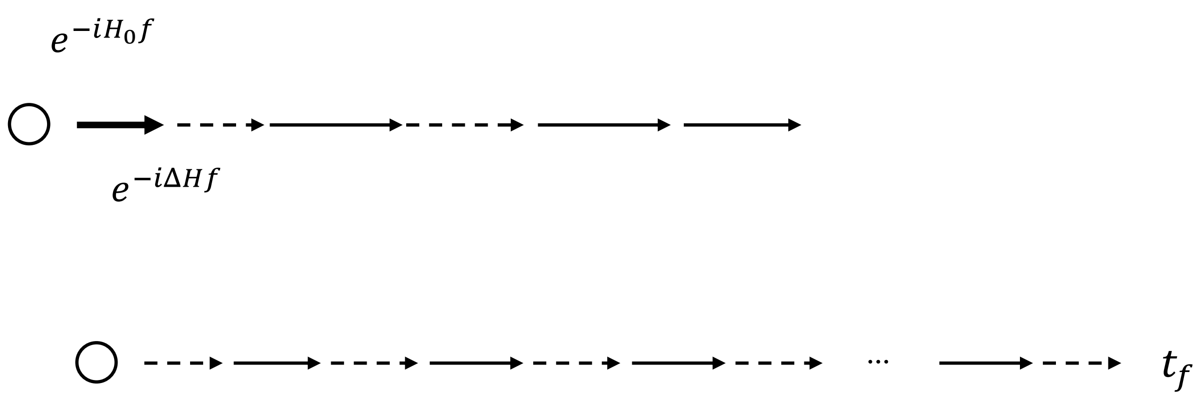


Figure 1: Top: Standard Bottom: Improved (symmetrized)

II Path Integrals

A Derivation

Feynman's PI = Continuous Limit of Trotterization in Coordinate Space

$$H = K + V \tag{18}$$

$$
\begin{aligned}
G(x_N, x_0, t) &= \lim_{\delta\to 0}\left[e^{-i\frac{\hat{V}}{2}\delta}e^{-i\frac{\hat{K}}{\delta}}e^{-i\frac{\hat{V}}{2}\delta}\right]^N \\
&= \lim_{\delta\to 0}\prod_{n=1}^{N}\int \langle x_n|\hat{G}(\delta)|x_{n-1}\rangle dx_n
\end{aligned}
\tag{19}
$$

$$
\begin{aligned}
G_\delta(x_n, x_{n-1}, \delta) &= \int dp\, \langle x_n|e^{-i\frac{\hat{V}}{2}\delta}|x_n\rangle\langle x_n|p\rangle e^{-ip^2\delta/(2m)}\langle p|x_{n-1}\rangle\langle x_{n-1}|e^{-i\frac{\hat{V}}{2}\delta}|x_{n-1}\rangle \\
&= \int dp\frac{1}{2\pi}\, e^{ip(x_n-x_{n-1})}e^{-i\frac{p^2}{2m}\delta}e^{-i\delta\left(V_n+V_{n-1}\right)\frac{1}{2}} \\
&= \left(\frac{m}{2\pi ik\delta}\right)^{1/2} e^{\frac{i(x_n-x_{n-1})^2}{2\delta}m-i\frac{1}{2}\left(V_n+V_{n-1}\right)\delta}
\end{aligned}
\tag{20}
$$

Here, we identify

$$
\lim_{\delta\to 0}\frac{x_n-x_{n-1}}{\delta} = \dot{x}_{n+\frac{1}{2}}
\tag{21}
$$

$$
\begin{aligned}
G(\delta) &= \left(\frac{m}{2\pi i\hbar\delta}\right)^{1/2}\exp\left[i\frac{\dot{x}^2_{n+\frac{1}{2}}}{2}m - i\overline{V}_{n+\frac{1}{2}}\delta\right] \\
&= \left(\frac{m}{2\pi i\hbar\delta}\right)^{1/2}\exp\left[S(x_n, x_{n-1}; \delta)\frac{i}{\hbar}\right],
\end{aligned}
\tag{22}
$$

$$
G(t) = \underbrace{\lim_{\delta\to 0}\prod\int dx_n\left[\frac{m}{2\pi\hbar\delta i}\right]}_{\downarrow} e^{i\sum S_n\frac{1}{\hbar}}
\tag{23}
$$

$$
= \qquad \int \mathcal{D}[x(t)] \quad e^{iS\frac{1}{\hbar}}
\tag{24}
$$

with $x_N = x_f$ and $S = \int \mathcal{L}dt$

B Equivalence of path integrals and Schrodinger equation

Free Particle (set $V = 0$)

$$
G_0 = \left[\frac{M}{2\pi i\hbar t_f}\right]^{1/2}\exp\left[i\frac{(x_f-x_0)^2}{2t_f h}m\right]
\tag{25}
$$

Markov property of G:

$$
\hat{G}\left(t_f\right) = \hat{G}\left(t_f - t_n\right)\hat{G}\left(t_2\right)
\tag{26}
$$

Apply to ψ:

$$
\psi(x,t) = \int d\delta x\, \hat{G}_0(\delta x, \delta t)\psi(x-\delta x, t-\delta t)
\tag{27}
$$

LHS:

$$
\psi(x,t) = \psi(x, t-\delta t) + \partial_t\psi(-\delta t)
\tag{28}
$$

RHS:

$$
\int \hat{G}(\delta x, \delta t)\left[\psi(x, t-\delta t) + \partial x\,\psi(-\delta x) + \frac{1}{2}\partial^2 4(-\delta x)^2\right]
\tag{29}
$$

$$
= \psi(x, t-\delta t) + \frac{1}{\partial}\partial^2\psi\underbrace{\int \hat{G}(\delta x, \delta t)\left(-\delta x^2\right)\, dy}_{-i\frac{\hbar^2}{m}\delta t}
\tag{30}
$$

$$
\therefore \quad i\dot{\psi} = -\frac{\hbar^2}{2m}\partial_x^2\psi = \frac{\hat{p}^2}{2m}\psi
\tag{31}
$$

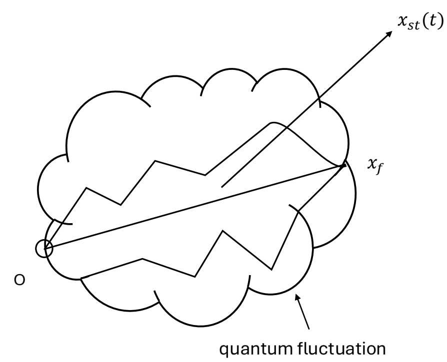
Easy to incoorperate V.
So, Feynman's PI = Schrödinger equation.

C Semi-classical approximation

Stationary solution

$$S[x(t)] = S_{st}(t) + \int \left.\frac{\delta S}{\delta x}\right|_{st} \delta x\, dt + \frac{1}{2}\iint \frac{\delta' S}{\delta x_1\,\delta x_2}\delta x_1\,\delta x_2\,dt \tag{32}$$

$$\frac{\delta S}{\delta x} = \underbrace{\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} - \frac{\partial \mathcal{L}}{\partial x}}_{\text{Euler-Lagrange Eq.}} = m\ddot{x} + \partial V = 0 \tag{33}$$



Second-order term represents quantum fluctuations.

$$G(+) = \int \mathcal{L}[x(t)]\exp\left[iS[x(t)]\frac{1}{\hbar}\right] \approx \left[\frac{1}{2\pi i\hbar}\left(-\frac{\partial^2 S_{st}}{\partial x(0)\partial x(t)}\right)\right]^{1/2}\exp(iS_{st}\hbar) \tag{34}$$

Here $S[x(t)]$ is a functional, but $S_{st}(x_0, x_f)$ is a function.

$$\frac{\partial^2 S}{\partial x_0 \partial x_f} = \frac{\partial p_0}{\partial x_f} = \left(\frac{\partial x_f}{\partial p_0}\right)^{-1} \Rightarrow \text{stability matrix} \tag{35}$$

When $\frac{\partial x_f}{\partial p_0} = 0$, G_{st} get divergence at the caustic, where two or more paths coalesce at a focal point.

III Imaginary-Time Formalism

A All thermodynamic quantities are determined by Q

$$Q = \operatorname{Tr}\tilde{G}(\beta) = \sum_n e^{-\beta\varepsilon_n}, \tag{36}$$

$$\tilde{G}(\beta) = \tilde{G}(-i\beta\hbar) = \sum_n e^{-\beta\varepsilon_n}\,|n\rangle\langle n| \tag{37}$$

Imaginary-time evolution:

$$-\partial_\tau \tilde{G} = \hat{H}\tilde{G}, \qquad 0 < \tau < \hbar\beta. \tag{38}$$

What is the path-integral representation?

$$\tilde{Q} = \oint \mathcal{L}[x(\tau)]\, e^{-\tilde{s}[x(\tau)]}. \tag{39}$$

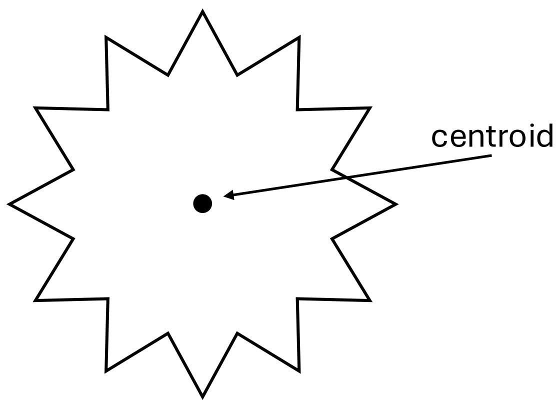
where

$$\tilde{S}(\tau) = \oint d\tau \left[\frac{m}{2}\dot{x}^2 + V(x,\tau)\right] \tag{40}$$

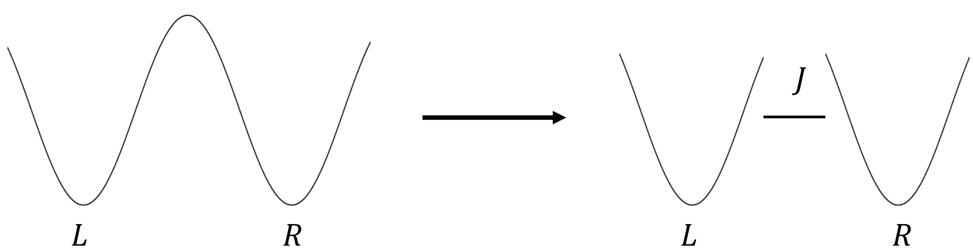
$$= \sum \left[\frac{m}{2}\left(\frac{x_n - x_{n-1}}{\delta}\right)^2 + V_n\right]\delta. \tag{41}$$

$$\delta = \frac{\hbar\beta}{N}. \tag{42}$$

Ring polymer



B Steepest-descent evaluation



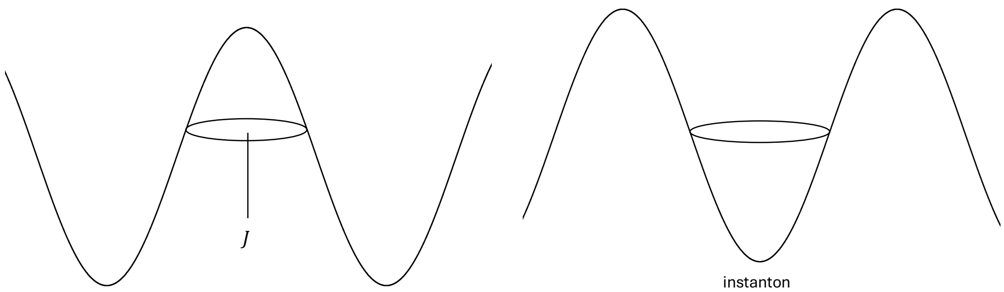
$$\begin{bmatrix} \varepsilon_R & J \\ J & \varepsilon_L \end{bmatrix} \overset{\text{resonance}}{\Rightarrow} 2J = \text{energy gap}. \tag{43}$$

How to evaluate J?

$$\begin{aligned} &S = \oint \frac{m\dot{x}^2}{2} + V\,d\tau, \\ &\frac{\delta S}{\delta x} = -m\ddot{x} + \nabla V = 0, \quad \text{or } m\ddot{x} = \nabla V, \\ &\qquad\qquad\qquad\qquad\quad\ \ \text{subject to period } \hbar\beta. \end{aligned} \tag{44}$$

So the tunneling splitting J becomes

$$J \propto e^{-\,\mathrm{S_{inst}}}/\hbar. \tag{45}$$



C Variational method

$$Q = \mathrm{Tr}\, e^{-\beta(\hat{H}_0 + \Delta \hat{H})} > Q_0 \left\langle e^{-\beta \Delta H} \right\rangle \tag{46}$$

where

$$\langle A \rangle_0 = \frac{\mathrm{Tr}\, A e^{-\beta H_0}}{T_r e^{-\beta H}}. \tag{47}$$

Free-energy minimization:

$$F < -kT \ln Q_0 + \langle \Delta H \rangle \tag{48}$$

Introduce the reference Hamiltonian

$$H_0 = \frac{1}{2} m\omega_0^2 (x - x_0)^2 \tag{49}$$

where ω_0 and x_0 are optimized.

Then

$$\begin{cases} \langle V' \rangle_0 = 0 \\ \langle V'' \rangle_0 = m\omega_0^2 \end{cases} \tag{50}$$

Application:

$$V = \frac{1}{2} m\omega^2 x^2 + cx^4. \tag{51}$$

$$\left\langle m\omega^2 + 2cx^2 \right\rangle_0 = m\omega_0^2. \tag{52}$$

Note:

$$\left\langle x^2 \right\rangle_2 = \frac{\hbar}{2m\omega_0} \,\mathrm{cth} \left(\frac{\hbar \omega_0 \beta}{2} \right), \tag{53}$$

$$\therefore \quad m\omega^2 + 12c \left\langle x^2 \right\rangle_0 = m\omega_0^2. \tag{54}$$

Alternatively,

$$\left\langle x^2 \right\rangle = \frac{\int e^{-\beta H} x^2 dx}{\int e^{-\beta H} dx} \tag{55}$$

$$H = \frac{m\omega^2}{2} x^2 + 6cx^2 \left\langle x^2 \right\rangle \tag{56}$$